Constraints in MySQL

What are Constraints?

Constraints are **rules applied to table columns** to ensure the **accuracy, reliability, and integrity of data** in a database.

Types of Constraints in MySQL

1. NOT NULL
2. UNIQUE
3. PRIMARY KEY
4. FOREIGN KEY
5. CHECK
6. DEFAULT

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1. NOT NULL Constraint

Definition

Ensures that a column **cannot have NULL (empty) values**.

Syntax

```
column_name datatype NOT NULL
```

Example

```
CREATE TABLE Student (
    id INT NOT NULL,
    name VARCHAR(50) NOT NULL
);
```

2. UNIQUE Constraint

Definition

Ensures that **all values in a column are different**.

Syntax

```
column_name datatype UNIQUE
```

Example

```
CREATE TABLE Users (
    email VARCHAR(100) UNIQUE
);
```

3. PRIMARY KEY Constraint

Definition

- Uniquely identifies each record
- Combination of **NOT NULL + UNIQUE**
- Only **one primary key per table**

Syntax

```
column_name datatype PRIMARY KEY
```

Example

```
CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50)
);
```

(Composite Primary Key)

A **composite primary key** is a **primary key made up of two or more columns**.
It is used when **a single column is not enough** to uniquely identify a record.

☞ **Combination of columns must be unique**, but individual columns **can repeat**.

Why Use a Composite Primary Key?

- To uniquely identify records using **multiple attributes**
- Common in **many-to-many relationships**

- Prevents **duplicate combinations**

Syntax

```
PRIMARY KEY (column1, column2)
```

Example 1: Student Course Enrollment

Table Creation

```
CREATE TABLE Enrollment (
    student_id INT,
    course_id INT,
    enroll_date DATE,
    PRIMARY KEY (student_id, course_id)
);
```

---

4. FOREIGN KEY Constraint

Definition

Creates a **relationship between two tables** and ensures **referential integrity**.

Syntax

```
FOREIGN KEY (column) REFERENCES table(column)
```

Example

```
CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);

CREATE TABLE Teacher (
    teacher_id INT PRIMARY KEY,
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);
```

(Composite Foreign Key)

A **composite foreign key** is a foreign key that **consists of two or more columns** in a child table which **together reference a composite primary key** (or unique key) in a parent table.

☞ The **number, order, and data types of columns must match**.

When is it Used?

- When the **parent table has a composite primary key**
- To maintain **referential integrity** for combinations of values
- Common in **many-to-many relationships**

Syntax

```
FOREIGN KEY (child_col1, child_col2)
REFERENCES parent_table (parent_col1, parent_col2)
```

Example 1: Course Enrollment System

Parent Table (Composite Primary Key)

```
CREATE TABLE Enrollment (
    student_id INT,
    course_id INT,
    enroll_date DATE,
    PRIMARY KEY (student_id, course_id)
);
```

Child Table (Composite Foreign Key)

```
CREATE TABLE Exam (
    student_id INT,
    course_id INT,
    marks INT,
    FOREIGN KEY (student_id, course_id)
    REFERENCES Enrollment (student_id, course_id)
);
```

5. CHECK Constraint

Definition

Ensures that values in a column **satisfy a specific condition**.

Syntax

```
CHECK (condition)
```

Example

```
CREATE TABLE Student_Age (
    age INT CHECK (age >= 18)
);
```

6. DEFAULT Constraint

Definition

Assigns a **default value** when no value is provided.

Syntax

```
column_name datatype DEFAULT value
```

Example

```
CREATE TABLE Orders (
    status VARCHAR(20) DEFAULT 'Pending'
);
```

Summary Table

Constraint	Purpose
NOT NULL	Prevents NULL values
UNIQUE	Prevents duplicate values
PRIMARY KEY	Unique + Not Null identifier
FOREIGN KEY	Links two tables
CHECK	Restricts values

Constraint	Purpose
DEFAULT	Assigns default value